A Report on

Synthesis and characterization of ZnSb intermetallic compound by mechanical alloying and its applications

Submitted by

Dungavath Sathish Naik

Roll No.: 123MM0678

B.Tech. – Metallurgical and Materials Engineering

Under the guidance of

Prof. Bankim Chandra Ray

SEMINAR AND TECHNICAL WRITING REPORT

Department of Metallurgical and Materials Engineering

NATIONAL INSTITUTE OF TECHNOLOGY, ROURKELA

Academic Year 2026–27

TABLE OF CONTENTS

Section	Title
1	Abstract
2	Extensive Background Literature
3	Explanations and Graphical Descriptions
4	Present Status of Commercialisation and Future Developments
5	Perspectives and Prospects
6	Remarks and Conclusions
7	References
A	Appendix: Visual Element Summary (Figures 1–16)

1. ABSTRACT

Zinc antimonide (ZnSb) is a binary intermetallic compound investigated as a p-type thermoelectric material for intermediate-temperature energy conversion (~200–400 °C). Thermoelectric materials convert waste heat directly into electrical energy without moving parts, which is relevant for improving energy efficiency in industrial and automotive systems. ZnSb is interesting because zinc and antimony are relatively abundant and less toxic than elements in competing thermoelectrics such as Bi_2Te_3 or PbTe. Its thermoelectric performance, measured by the dimensionless figure of merit $ZT = S^2\sigma T/\kappa$, depends strongly on phase purity, composition, grain size and defect concentration.

Mechanical alloying (MA) is a solid-state powder-processing technique in which elemental powders are subjected to repeated high-energy impacts. The repeated deformation, cold welding, fracture and rewelding refine the microstructure, create lattice defects and fresh interfaces, and shorten diffusion distances—promoting solid-state interdiffusion and nanocrystalline ZnSb formation without melting. Published studies have reported complete ZnSb formation within 3–4 h under optimized planetary milling conditions [1, 2].

In the present work, elemental Zn and Sb powders were mechanically ball milled for 25 h, followed by annealing at 400 °C for 2 h. The processed sample was characterized by X-ray diffraction (XRD). The obtained pattern shows several distinct crystalline reflections, but reliable phase identification requires careful comparison with standard reference data, especially because the Zn–Sb system is composition-sensitive and zinc can evaporate during elevated-temperature treatment [5].

This report reviews the scientific background of ZnSb, explains the mechanism of mechanical alloying, discusses previous research, presents the experimental XRD observation, and evaluates the potential of ZnSb for thermoelectric energy conversion. Major processing challenges—zinc loss, phase competition, thermal instability and reproducibility—are discussed alongside future directions.

Keywords: ZnSb; zinc antimonide; mechanical alloying; ball milling; intermetallic compound; thermoelectric material; X-ray diffraction; nanocrystalline; waste heat recovery.

Graphical Abstract

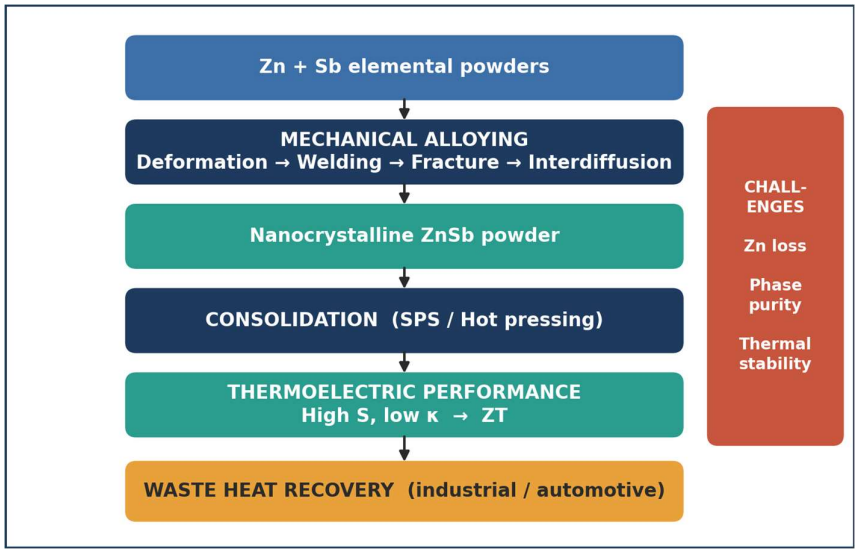


Fig. 1. Graphical abstract: processing–structure–property–application chain for ZnSb by mechanical alloying. Conceptual schematic prepared for the present report.

2. EXTENSIVE BACKGROUND LITERATURE

2.1 Intermetallic Compounds and ZnSb

An intermetallic compound is an ordered solid-state phase formed between metallic or semimetallic elements with a defined composition and crystal structure. Unlike a random solid solution—where solute atoms randomly substitute for solvent atoms—an intermetallic has specific lattice sites occupied by each element in an ordered arrangement. This ordering controls bonding, electronic structure and defect behaviour, producing properties quite different from either constituent element.

Zinc antimonide (ZnSb) is a binary intermetallic at 1:1 atomic Zn:Sb ratio. It crystallizes in an orthorhombic structure (space group Pbca) at room temperature and is reported as a p-type semiconductor investigated for thermoelectric energy conversion [1, 2]. Several features make ZnSb attractive:

- **Elemental abundance:** Zn and Sb are more abundant and less expensive than Te (needed for Bi_2Te_3) or Pb (used in PbTe) [1, 2].
- **Lower toxicity** compared to lead-based thermoelectrics.
- **Intermediate-temperature operating range** (~200–400 °C), matching many industrial waste-heat sources [2].

However, ZnSb presents important processing difficulties. Zinc has appreciable vapour pressure at moderate temperatures; Song et al. [5] reported ZnSb decomposition beginning at approximately 300 °C as zinc evaporated and metallic Sb precipitated. Small composition shifts can generate unwanted phases because the Zn–Sb system contains several neighbouring intermetallics ($Zn_{13}Sb_{10}$, Zn_3Sb_2, etc.) [6]. Phase purity is therefore a central technical challenge.

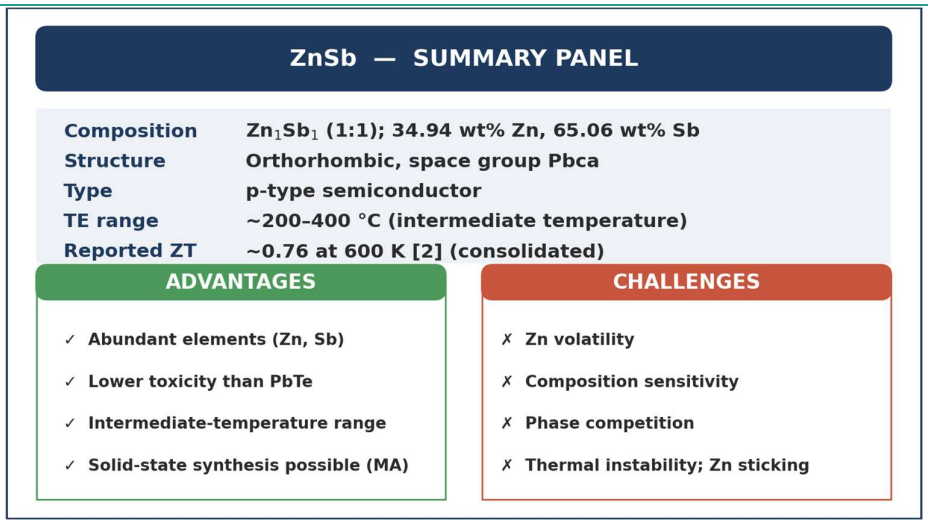


Fig. 2. Summary panel for ZnSb. Compiled from Refs. [1, 2, 5, 6].

2.2 The Zn–Sb Phase Diagram

Lo et al. [6] revised the Zn–Sb phase diagram using high-temperature diffraction and elemental analysis. The important features for ZnSb synthesis are:

- **ZnSb** (target phase): a line compound at ~50 at.% Sb with a narrow stability range. Small composition deviations push the system into two-phase fields.
- **$Zn_{13}Sb_{10}$** (often written Zn_4Sb_3): a Zn-rich phase at ~43.5 at.% Sb that can accommodate slight Zn deficiency [6].
- **Zn_3Sb_2 and Zn_9Sb_7**: additional phases stable at high temperature [6].
- **Terminal phases**: elemental Zn (m.p. 419.5 °C) and Sb (m.p. 630.6 °C).

The practical consequence: if zinc is lost during milling (sticking) or annealing (evaporation), the effective composition shifts, potentially producing Sb-rich secondary phases or $Zn_{13}Sb_{10}$ alongside or instead of ZnSb. Lee and Lin [7] showed that even 0.6 at.% Zn excess strongly changed phase formation in the related Zn_4Sb_3 system.

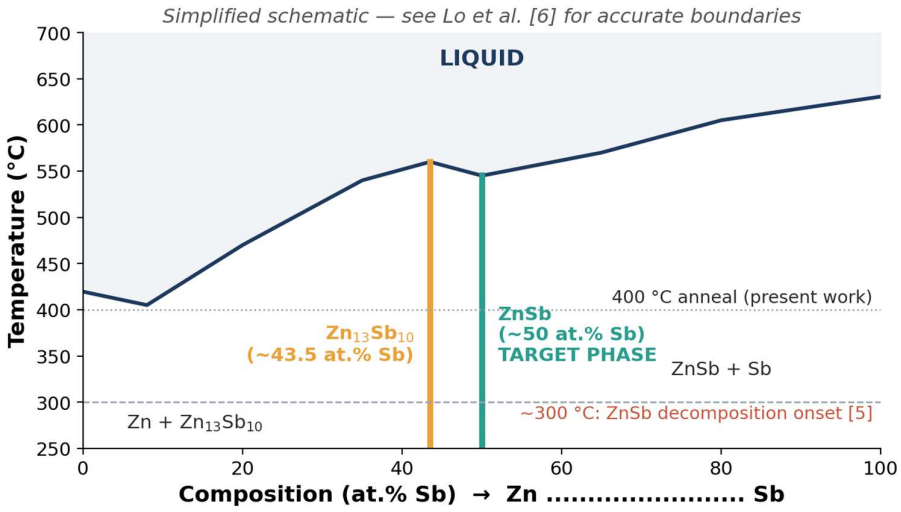


Fig. 3. Schematic Zn–Sb binary phase diagram. Simplified; for accurate phase boundaries see Lo et al. [6].

2.3 Thermoelectric Fundamentals

Thermoelectric materials convert a temperature difference into electrical energy via the Seebeck effect. When a temperature gradient is maintained across a semiconductor, charge carriers at the hot end have more thermal energy and diffuse toward the cold end, creating a voltage. The Seebeck coefficient S (μV/K) quantifies this voltage per unit temperature difference. The dimensionless figure of merit is:

$$ZT = S^2\sigma T / \kappa$$

where σ is electrical conductivity, T is absolute temperature and κ is total thermal conductivity. The numerator $S^2\sigma$ is the **power factor** (PF). Total thermal conductivity has two components:

$$\kappa = \kappa_L + \kappa_e$$

where κ_L is the lattice (phonon) contribution and κ_e is the electronic contribution, linked to σ through the Wiedemann–Franz law ($\kappa_e = L\sigma T$, where L is the Lorenz number).

High ZT requires simultaneously high S, high σ and low κ—a difficult optimization because these properties are interdependent. Increasing carrier concentration raises σ but lowers S and increases κ_e. The main independent variable for improvement is reducing κ_L through microstructural engineering—specifically grain refinement, which is inherently produced by mechanical alloying.

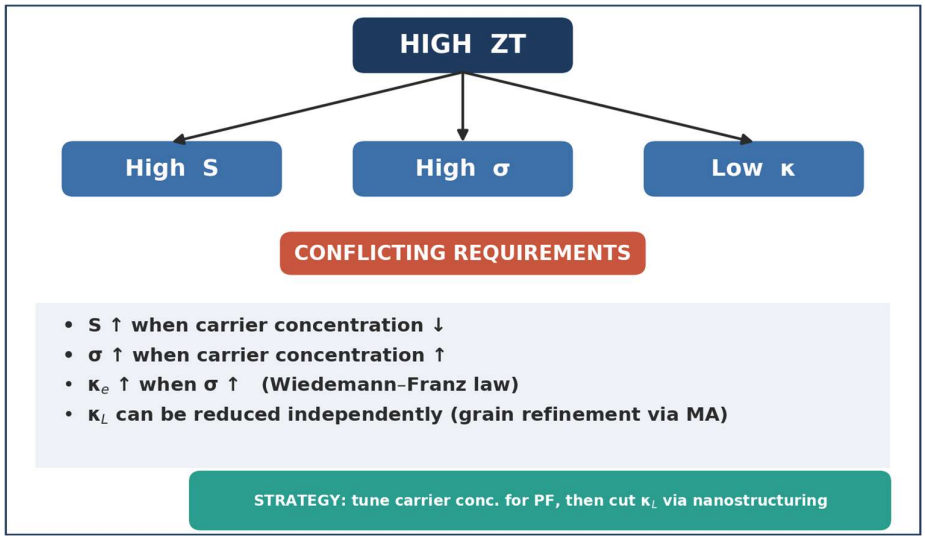


Fig. 4. The thermoelectric optimization trade-off. Schematic prepared for the present report.

2.4 Mechanical Alloying: Principles

Mechanical alloying is a non-equilibrium solid-state powder-processing technique in which elemental powders in a vial with milling balls undergo repeated high-velocity collisions. Suryanarayana [8] provides a comprehensive review. The repeated impacts produce:

1. **Plastic deformation:** Ductile particles flatten and work-harden. Zn (ductile, HCP) predominantly deforms.
2. **Fracture:** Brittle particles (or work-hardened ones) break. Sb (comparatively brittle) tends to fracture.
3. **Cold welding:** Fresh surfaces created by deformation/fracture bond under pressure, forming composite particles with alternating Zn-rich and Sb-rich layers.
4. **Rewelding and refinement:** Repeated cycles progressively refine the internal lamellar spacing to nanometres.
5. **Defect accumulation:** Dislocations (potentially 10^{11}–10^{12} cm^{-2}), grain boundaries, stacking faults and other defects accumulate, raising stored energy and providing short-circuit diffusion paths [8].

6. **Solid-state interdiffusion:** With nanometre-scale diffusion distances and high defect density, interdiffusion between Zn and Sb accelerates—even at room temperature or the modest temperatures generated by milling—leading to ZnSb nucleation and growth.

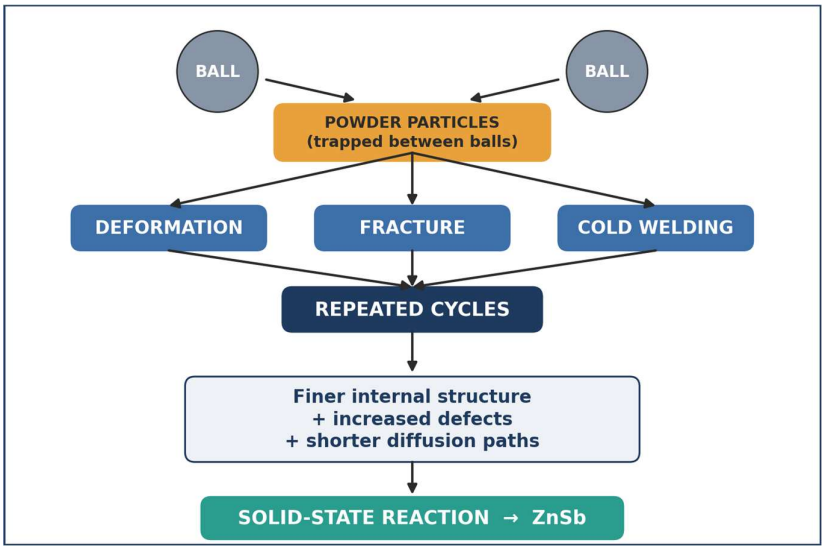


Fig. 5. Ball–powder–ball interaction during mechanical alloying. Based on Suryanarayana [8].

2.5 Mechanical Alloying of Zn–Sb: Formation Mechanism

The Zn–Sb system is a ductile–brittle combination [8]. Based on the literature [1, 2, 3], ZnSb formation proceeds through distinct stages (Fig. 6).

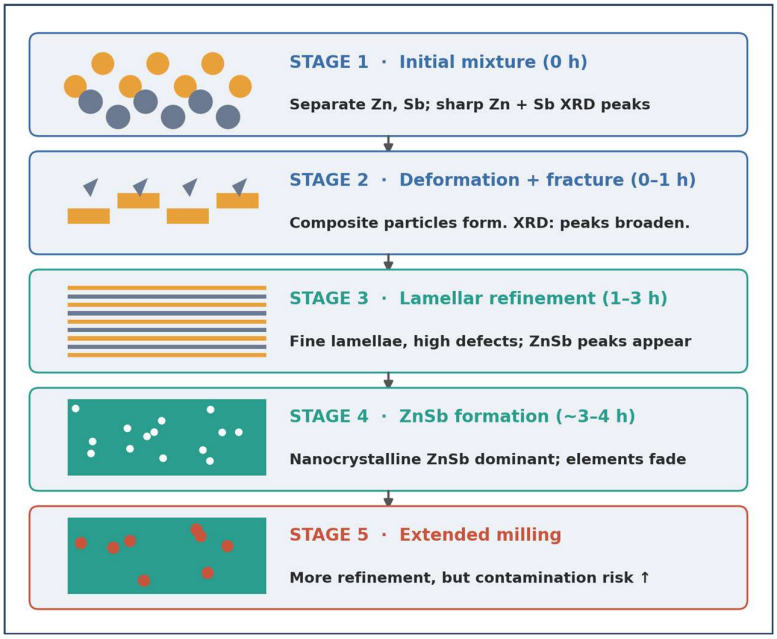


Fig. 6. Five-stage ZnSb formation mechanism during MA. Schematic based on Refs. [1, 2, 3, 8].

The key physical insight is that ZnSb forms without melting because: (a) lamellar refinement reduces diffusion distances to nanometres, (b) extreme defect density provides short-circuit diffusion paths where grain-boundary and dislocation-pipe diffusion are orders of magnitude faster than lattice diffusion, and (c) stored energy from plastic deformation lowers the thermodynamic barrier for intermetallic nucleation [8].

2.6 Previous ZnSb Synthesis Studies

Table 1. Comparison of major Zn–Sb synthesis studies. "NR" = not reported. BPR = ball-to-powder ratio; MG = mechanical grinding; HP = hot pressing.

Study	Target	Mill; speed; BPR; atm.	Time for ZnSb	Composition; crystallite size	Consolidation; ZT	Key observation
Pothin et al. (2016) [1]	ZnSb (direct MA)	Fritsch P5; 400 rpm; 20:1; Ar	~3 h	Slight Zn excess; ~40 nm	SPS; characterized	Fast formation; Zn excess + heat control needed
Balasubramanian et al. (2021) [2]	ZnSb (direct MA)	Fritsch P5; 350 rpm; 10:1; Ar	~4 h	Stoichiometric 1:1; nanocrystalline	Consolidated; ZT ~0.76 at 600 K	Phase kinetics mapped; best moderate-energy baseline
Okamura et al. (2010) [3]	ZnSb (MG of pre-alloyed)	Planetary; ≤180 rpm; NR; Ar	100 h (grinding)	Pre-alloyed; fine-grained	HP at 673 K; lower κ	Zn sticking identified as major obstacle
Lee & Lin (2018) [7]	β-Zn_4Sb_3	SPEX 8000D; N/A; ~5:1; Ar	5 h	0.6 at.% Zn-rich; NR	Multiple routes; characterized	Small Zn excess critically changes phase constitution

Key observations from the literature:

1. **Planetary mills dominate** the successful direct-MA studies, with Fritsch P5 used by both [1] and [2].
2. **3–4 h suffices** under high-energy conditions (350–400 rpm, BPR 10–20:1).
3. **Composition sensitivity is critical**—even 0.6 at.% Zn change alters the outcome [7].
4. **Thermal treatment is risky**: ZnSb decomposes above ~300 °C [5]; consolidation at elevated temperature caused Zn evaporation and phase transformation in [7].

2.7 Research Gap

The literature establishes that ZnSb can be synthesized by MA, but important gaps remain:

- **Zn retention quantification:** No systematic study of how much Zn is lost during milling versus heating versus consolidation.
- **Process transferability:** Conditions optimized on one mill may not transfer directly to another.
- **Post-milling thermal treatment window:** The interplay between beneficial effects (strain relief, crystallinity) and harmful effects (Zn loss, decomposition) of annealing on MA-produced ZnSb is not well defined.
- **Long-term thermal stability** of nanocrystalline ZnSb at operating temperatures is insufficiently studied.
- **Contamination assessment** (W/Co from WC media) is rarely quantitative.

The present work contributes an experimental case with 25 h milling (substantially longer than the 3–4 h benchmark) and 400 °C annealing (above the ~300 °C decomposition onset), providing XRD data for evaluation of the combined effect.

3. EXPLANATIONS AND GRAPHICAL DESCRIPTIONS

3.1 Experimental Route

The actual processing route in the present work is summarised in Fig. 7.

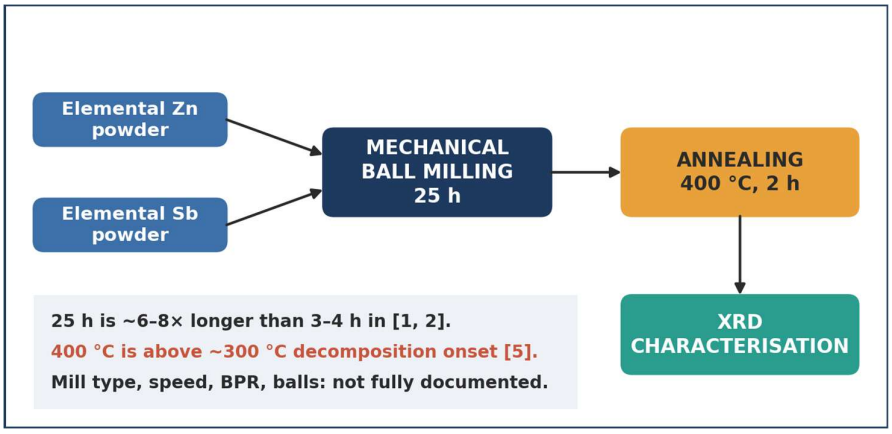


Fig. 7. Experimental processing route used in the present work. Source: experimental information from the present project.

This route differs from published benchmarks in two important ways. First, the milling time is substantially longer—25 h versus 3–4 h. Without intermediate sampling, it is unknown at what point phase formation was actually complete. Second, the annealing temperature exceeds the decomposition onset reported by Song et al. [5], raising the possibility of Zn loss during thermal treatment. These observations frame the XRD interpretation below.

3.2 Effect of Milling Parameters

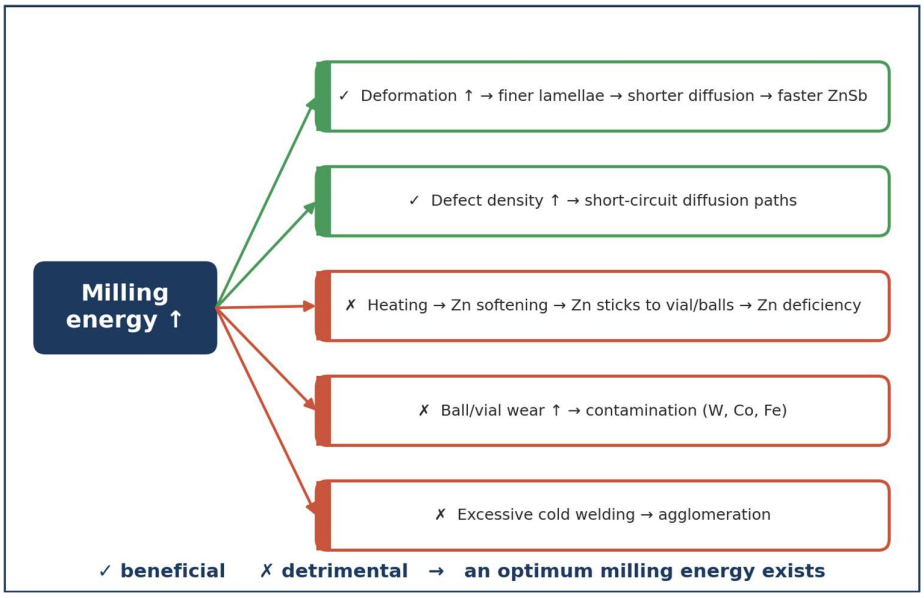


Fig. 8. Beneficial and detrimental effects of increasing milling energy. Based on Refs. [1, 2, 3, 8].

Table 2. MA parameters and their influence on ZnSb formation.

Parameter	Effect on ZnSb	Literature basis
Speed ↑	Faster reaction; but T rise, Zn sticking, contamination ↑	[1, 2, 8]
BPR ↑	More impacts/g; faster reaction; contamination ↑	[1, 2, 8]
Time ↑	More complete reaction; beyond optimum → contamination	[2]
Ar atmosphere	Reduces oxidation	[1, 2, 3, 7]

Parameter	Effect on ZnSb	Literature basis
WC media	High energy transfer; W/Co contamination risk	[1, 2]
Zn excess	Compensates for Zn loss; but may produce Zn-rich phases	[1, 7]
Duty cycle (rest periods)	Limits T rise; reduces Zn sticking	[1]

3.3 Post-Milling Annealing: Competing Effects

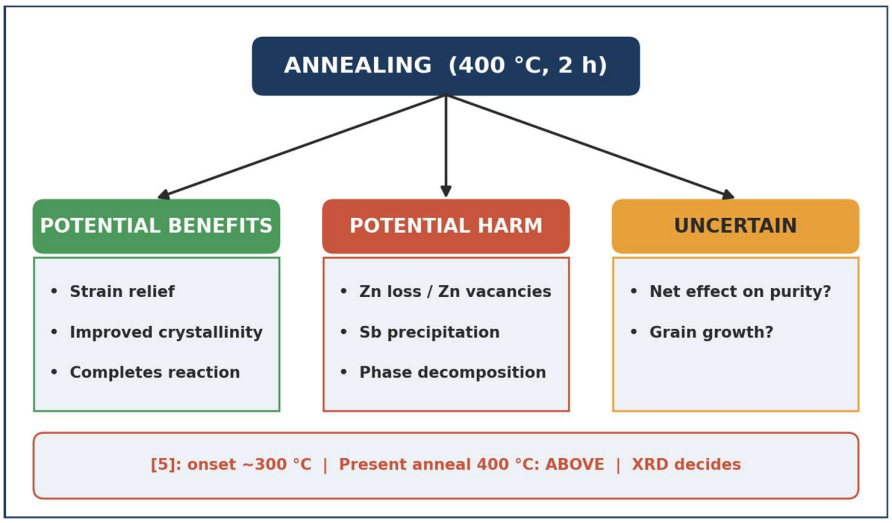


Fig. 9. Competing effects of post-milling annealing on MA ZnSb. The 400 °C treatment is above the ~300 °C decomposition onset [5]. Schematic prepared for the present report.

3.4 X-Ray Diffraction and Phase Identification

XRD is the primary characterization technique used here. When monochromatic X-rays interact with a crystalline material, constructive interference occurs at angles satisfying Bragg's law:

$$n\lambda = 2d\sin\theta$$

Each crystalline phase produces a characteristic set of reflections that acts as a fingerprint. For mechanically alloyed powders, peak broadening also carries information about crystallite size (Scherrer equation) and microstrain (separable using the Williamson–Hall method):

$$D = K\lambda / (\beta\cos\theta) \qquad \text{(Scherrer)}$$

$$\beta\cos\theta = K\lambda / D + 4\varepsilon\sin\theta \qquad \text{(Williamson–Hall)}$$

The Scherrer equation alone tends to underestimate crystallite size because it attributes all broadening to size rather than strain [8]. In the Zn–Sb system, XRD is especially critical because multiple phases can coexist (Zn, Sb, ZnSb, $Zn_{13}Sb_{10}$, ZnO, Sb_2O_3) and some reflections may overlap. Phase identification requires matching peak positions, relative intensities and the complete expected reflection set—not just the strongest peak.

3.5 Experimental XRD Pattern and Interpretation

The XRD pattern obtained after 25 h milling + 400 °C/2 h annealing is shown in Fig. 10. The pattern extends approximately from 10° to 90° 2θ with several distinct reflections.

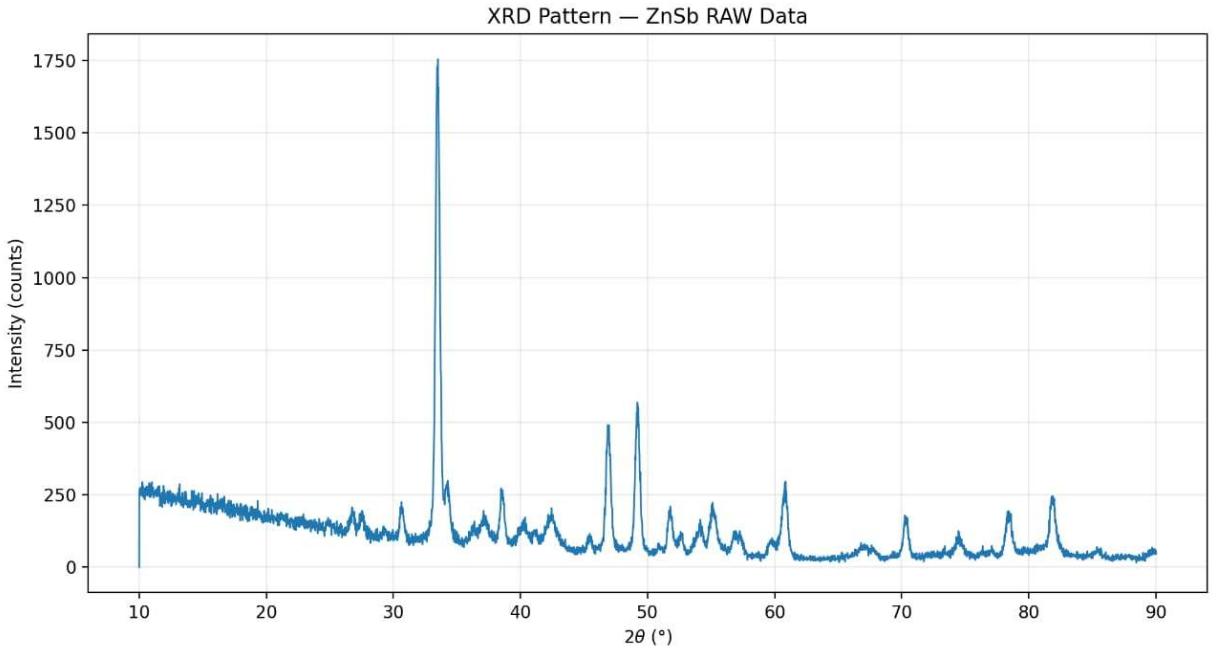


Fig. 10. XRD pattern of the Zn–Sb sample after 25 h milling + 400 °C/2 h annealing. Source: experimental data from the present work.

Table 3. Approximate prominent features in the experimental XRD pattern. Values are approximate 2θ readings from the plotted image, not final phase assignments. Exact positions should be obtained from the raw data file.

Approx. 2θ region	Relative prominence	Observation
33–34°	Very high	Strongest visible reflection
~47°	High	Distinct reflection
~49°	High	Distinct reflection
~61°	Moderate	Higher-angle reflection
~70°	Moderate	Distinct reflection
~78°	Moderate	Distinct reflection
~82°	Moderate	Distinct reflection

Interpretation

The pattern clearly shows crystalline material with multiple well-defined reflections. However, definitive phase identification requires a systematic comparison with standard ICDD/PDF data for ZnSb (Pbca), Sb, Zn, $Zn_{13}Sb_{10}$ and possible oxides. Several considerations are important:

1. The strong peak near 33–34° is consistent with ZnSb reflections, but Sb also has reflections in nearby regions. Both must be checked against reference data.
2. Since the annealing temperature (400 °C) exceeds the ~300 °C decomposition onset [5], partial Zn loss and free Sb precipitation cannot be excluded without confirming that no Sb-only peaks are present.
3. After 25 h of milling, contamination from the milling media could contribute additional peaks.
4. A definitive claim of single-phase ZnSb cannot be made from the image alone. Peak-by-peak matching using the raw data file is the essential next analysis step.

Honest assessment

The XRD pattern provides evidence consistent with the possibility that ZnSb has formed. The observed reflections should be treated as experimental observations at this stage, pending proper reference matching.

3.6 Structure–Property Relationship

The connection between MA microstructure and thermoelectric properties operates through two parallel pathways (Fig. 11).

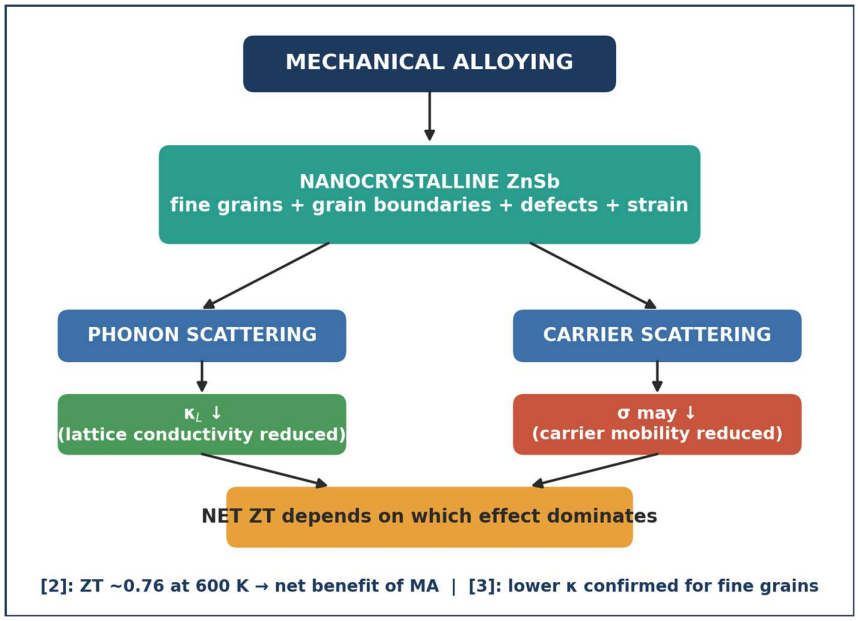


Fig. 11. Processing–structure–property relationship in MA ZnSb. Based on Refs. [2, 3, 5, 8].

Grain boundaries scatter phonons (reducing κ_L) but also scatter charge carriers (potentially reducing σ). Whether ZT improves depends on the relative magnitudes of these competing effects. Balasubramanian et al. [2] achieved ZT ~0.76 at 600 K, suggesting that for their MA conditions, the κ_L reduction outweighed the σ reduction. Okamura et al. [3] directly confirmed that fine-grained ZnSb has lower thermal conductivity than coarser material.

Additionally, zinc vacancies formed during thermal treatment [5] modify hole concentration, affecting both S and σ. This means defects are not merely processing imperfections—they actively control the thermoelectric response.

3.7 Consolidation

For thermoelectric applications, MA powder must be consolidated into a dense solid. Two main methods have been used for ZnSb:

- **Spark Plasma Sintering (SPS):** Rapid heating (minutes) with pulsed current + pressure; limits grain growth. Used by Pothin et al. [1].
- **Hot Pressing (HP):** Steady pressure + heating; longer exposure. Used by Okamura et al. [3] at 673 K.

Both methods risk Zn loss at elevated temperatures. SPS is preferred for ZnSb because its rapid heating minimizes thermal exposure. Consolidation was not performed in the present work.

3.8 Applications

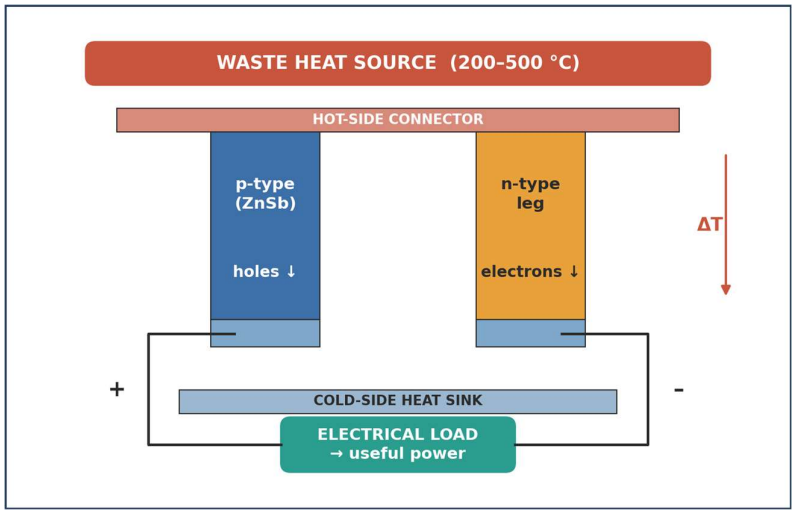


Fig. 12. Thermoelectric generator with ZnSb as p-type leg. Schematic prepared for the present report.

Table 4. Potential ZnSb thermoelectric applications. All are at research or concept stage; no commercial ZnSb TEG products were identified in the reviewed literature.

Application area	Heat source T	Status
Industrial waste heat recovery	200–500 °C	Research / early development
Automotive exhaust	200–600 °C	Research
Process industries (cement, steel)	200–500 °C	Research / concept
Remote off-grid power	Various	Concept

4. PRESENT STATUS OF COMMERCIALISATION AND FUTURE DEVELOPMENTS

4.1 Current Status

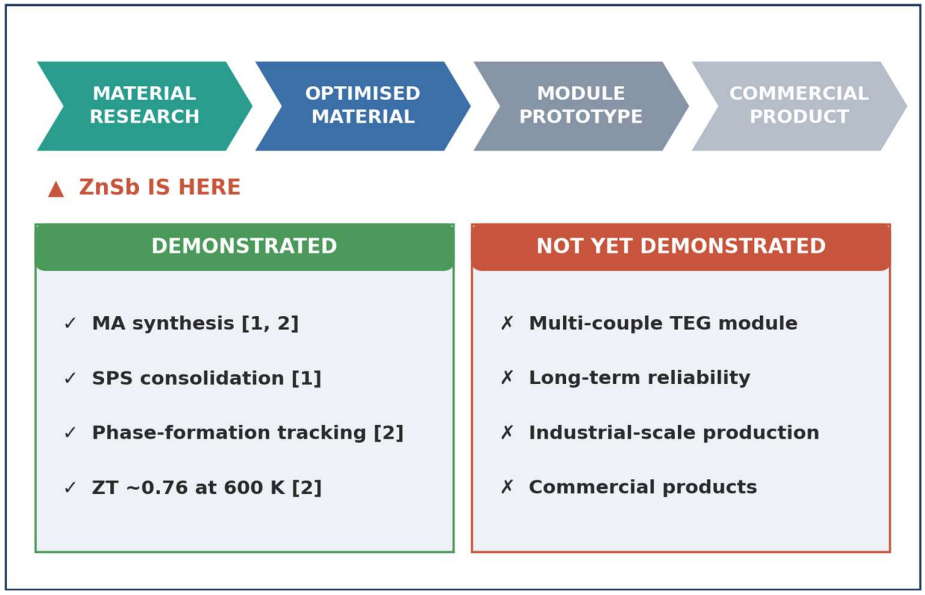


Fig. 13. Technology readiness of ZnSb. Schematic prepared for the present report.

4.2 Commercialisation Barriers

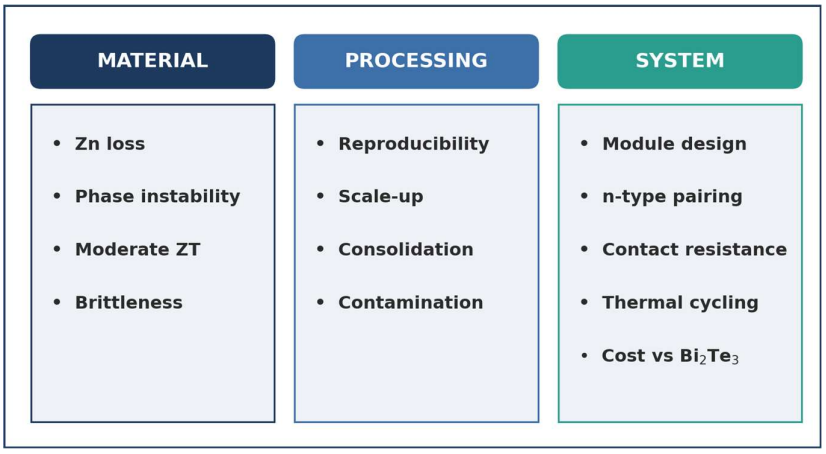


Fig. 14. Barrier categories for ZnSb commercialisation.

Table 5. Barriers and mitigation strategies.

Barrier	Technical issue	Possible mitigation
Zn loss	Composition shift → secondary phases	Inert/sealed processing; minimal thermal exposure [5]
Phase instability	Decomposition above ~300 °C	Operate below threshold; protective design
Moderate ZT	~0.5–0.8 for undoped ZnSb	Doping; nanocomposites; band engineering
Reproducibility	Mill-dependent results	Standardized protocols; robust process windows
Scale-up	Lab batches ~5–10 g	Larger mills; continuous processing (not yet demonstrated)

4.3 Future Developments

Realistic future research directions based on the identified gaps:

1. **Systematic MA parameter optimization** across different mill types to establish transferable process windows.
2. **Quantitative Zn retention tracking** (e.g., ICP analysis) through each processing step.
3. **Doping and alloying** to optimize carrier concentration and ZT.
4. **Nanocomposite approaches**—nanoscale inclusions for additional phonon scattering.
5. **Rapid low-temperature consolidation** to minimize Zn loss and grain growth.
6. **Long-term thermal aging studies** at operating temperatures.
7. **TEG module demonstration** with ZnSb as p-type leg paired with a compatible n-type material.

5. PERSPECTIVES AND PROSPECTS

5.1 My Perspective on the Future of ZnSb

1. I believe the future of ZnSb as a thermoelectric material will depend not only on its efficiency in converting heat to electricity, but also on its reliability in practical operating conditions. A material may look promising in the lab, but may still have problems when it is made in larger quantities, or heated and cooled over and over again. Therefore, ZnSb should be viewed from the perspective of materials science as well as engineering application.
2. ZnSb shows an interesting interplay between its composition, crystal structure and thermoelectric behavior. The phase formation, grain size, defects and elemental distribution depend on the processing conditions, which in turn affect the electrical conductivity and heat transport. Thus, the overall suitability should be evaluated based on several characterization techniques and properties.

5.2 Moving from Laboratory Synthesis to Practical Use

1. In my opinion research should move forward in defined steps: making reproducible powder controlling how the material is put together measuring transport properties testing stability and finally evaluating prototypes. Going step by step makes it easier to spot what's holding things back of changing too many processing factors at once.s
2. From an engineering point of view improving one part of a material doesn't mean the whole thing gets better. For example making grain size smaller can increase phonon scattering and lower conductivity from the lattice.. Smaller grains also scatter electrons, which can reduce electrical conductivity. So the real effect must be found through experiments, not assumptions.
3. I think future studies on ZnSb should report the Seebeck coefficient, electrical conductivity and thermal conductivity over the temperature range. Researchers should also talk about the trade-offs, between these properties of only showing the highest ZT value they measured. That way, comparing ways of making the material becomes more accurate and realistic.5.3 Reliability and Environmental Conditions
4. Another direction I consider important is understanding how ZnSb behaves during prolonged operation. Thermoelectric devices may experience repeated temperature changes, mechanical stresses, and contact degradation. A material that performs well during an initial measurement may not maintain that performance throughout its service life.
5. Future investigations could examine phase stability, changes in composition, oxidation sensitivity, dimensional stability, and the effect of repeated thermal cycling. Where appropriate, researchers could also evaluate protective coatings, suitable encapsulation, and contact materials. These are proposed research directions; their effectiveness for a particular ZnSb composition would need to be demonstrated.

5.3 My Proposed Research Direction

If I were to continue this investigation, I would organize the next stage around three questions:

1. **Can the desired ZnSb phase and composition be produced consistently?** This requires verified phase identification, composition measurements, and repeat experiments.
2. **How does processing influence transport properties?** Controlled variations in milling and annealing conditions could be related to microstructure and measured thermoelectric properties.
3. **Does the material remain useful under practical conditions?** Consolidated specimens should be assessed for mechanical integrity, thermal stability, and repeatability before device-level development.

I would avoid changing many parameters at once because doing so would make it difficult to identify which factor caused an observed improvement or deterioration. A controlled experimental plan would provide clearer evidence and help reduce unnecessary trials.

5.4 Overall Outlook

My overall view is that the strongest future direction for ZnSb research is to connect materials processing with measurable performance and practical reliability. The objective should not simply be to obtain a fine powder or an attractive diffraction pattern, but to establish a reproducible relationship between processing conditions, structure, transport properties, and operating stability.

I believe ZnSb deserves further investigation where its measured performance and processing requirements offer a meaningful advantage over competing materials. At the same time, its practical value should be judged objectively, including its limitations. This balanced approach would help determine whether further optimization, alternative processing conditions, or a different thermoelectric material is the most appropriate next step.

6. REMARKS AND CONCLUSIONS

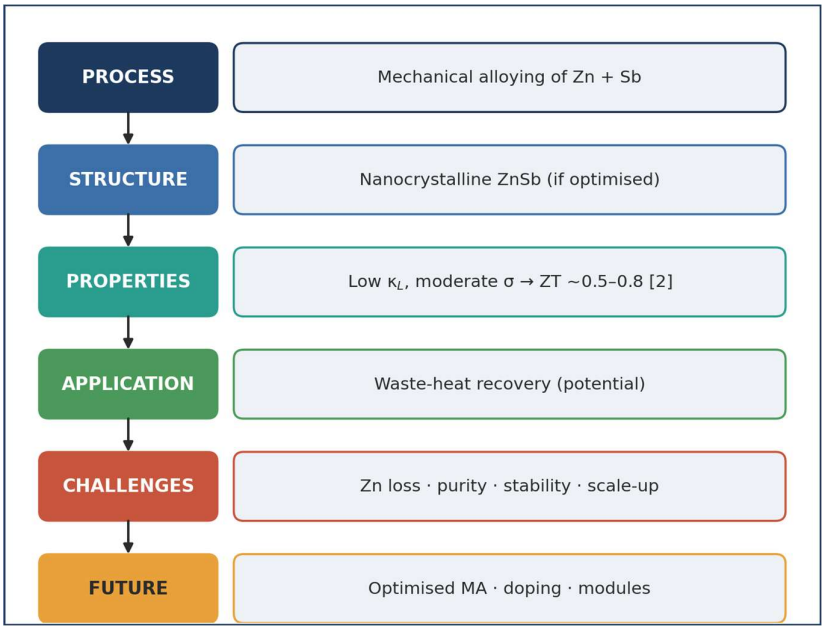


Fig. 16. Final summary of the processing–structure–property–application–challenges–future chain.

The main conclusions from this work are:

1. **ZnSb** is a binary Zn–Sb intermetallic (orthorhombic, Pbca, p-type) investigated for intermediate-temperature thermoelectric energy conversion where it offers advantages in elemental abundance and lower toxicity compared to Bi_2Te_3 and PbTe.
2. **Mechanical alloying** provides a solid-state route to nanocrystalline ZnSb through repeated deformation, cold welding, fracture and rewelding of Zn and Sb powders, which refine the microstructure, create defects and shorten diffusion distances, enabling intermetallic formation without melting. Published studies report complete ZnSb formation within 3–4 h under optimized high-energy planetary milling [1, 2].
3. **The Zn–Sb system is processing-sensitive.** Zinc sticking, zinc evaporation, composition shifts and phase competition are real obstacles. The phase diagram contains multiple neighbouring compounds with narrow stability fields [6], and ZnSb can decompose above ~300 °C through Zn loss [5].
4. **In the present work**, Zn and Sb powders were milled for 25 h and annealed at 400 °C for 2 h. The XRD pattern shows multiple crystalline reflections consistent with the presence of crystalline Zn–Sb phases. A definitive claim of single-phase ZnSb requires peak-by-peak comparison with standard reference data, which is the essential next step.
5. **Grain refinement through MA** can reduce lattice thermal conductivity via phonon-boundary scattering, contributing to improved ZT. Balasubramanian et al. [2] achieved ZT ~0.76 at 600 K for MA-synthesized ZnSb after consolidation.
6. **Practical thermoelectric application** of ZnSb requires solving the zinc retention problem, demonstrating long-term phase stability at operating temperatures, developing reproducible processing and consolidation routes, and integrating ZnSb into multi-couple thermoelectric modules. At present, ZnSb remains at the laboratory research stage.
7. **Future work** should focus on time-resolved XRD during milling to identify the actual completion point, annealing below the decomposition threshold in sealed/inert conditions, quantitative Zn retention analysis, doping for carrier optimization, and eventually module-level demonstration.

7. REFERENCES

[1] R. Pothin, R. M. Ayral, A. Berche, D. Granier, F. Rouessac, and P. Jund, "Preparation and properties of ZnSb thermoelectric material through mechanical-alloying and Spark Plasma Sintering," *Chemical Engineering Journal*, vol. 299, pp. 126–134, 2016. DOI: 10.1016/j.cej.2016.04.063.

[2] P. Balasubramanian, M. Battabyal, A. C. Bose, and R. Gopalan, "Effect of ball-milling on the phase formation and enhanced thermoelectric properties in zinc antimonides," *Materials Science and Engineering: B*, vol. 271, 115274, 2021. DOI: 10.1016/j.mseb.2021.115274.

[3] C. Okamura, T. Ueda, and K. Hasezaki, "Preparation of Single-Phase ZnSb Thermoelectric Materials Using a Mechanical Grinding Process," *Materials Transactions*, vol. 51, no. 5, pp. 860–862, 2010. DOI: 10.2320/matertrans.MH200902.

[4] R. Pothin, R.-M. Ayral, A. Berche, F. Rouessac, and P. Jund, "Interest of the differential thermal analysis on the characterization of a thermoelectric material: ZnSb," *Journal of Alloys and Compounds*, vol. 657, pp. 358–365, 2016. DOI: 10.1016/j.jallcom.2015.10.118.

[5] X. Song, M. Schrade, N. Masó, and T. G. Finstad, "Zn vacancy formation, Zn evaporation and decomposition of ZnSb at elevated temperatures: Influence on the microstructure and the electrical properties," *Journal of Alloys and Compounds*, vol. 710, pp. 762–770, 2017. DOI: 10.1016/j.jallcom.2017.03.339.

[6] C.-W. T. Lo, V. Svitlyk, D. Chernyshov, and Y. Mozharivskyj, "The updated Zn-Sb phase diagram. How to make pure $Zn_{13}Sb_{10}$ ('Zn_4Sb_3')," *Dalton Transactions*, vol. 47, pp. 11512–11520, 2018. DOI: 10.1039/C8DT02521E.

[7] P.-Y. Lee and P.-H. Lin, "Evolution of Thermoelectric Properties of Zn_4Sb_3 Prepared by Mechanical Alloying and Different Consolidation Routes," *Energies*, vol. 11, no. 5, 1200, 2018. DOI: 10.3390/en11051200.

[8] C. Suryanarayana, "Mechanical alloying and milling," *Progress in Materials Science*, vol. 46, pp. 1–184, 2001. DOI: 10.1016/S0079-6425(99)00010-9.

[9] D. M. Trichês, S. M. Souza, J. C. de Lima, T. A. Grandi, C. E. M. Campos, A. Polian, J. P. Itié, F. Baudelet, and J. C. Chervin, "High-pressure phase transformation of nanometric ZnSb prepared by mechanical alloying," *Journal of Applied Physics*, vol. 106, 013509, 2009. DOI: 10.1063/1.3158523.